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Forecasting Realized Volatility: The Choice of Window Size

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ABSTRACT

Different window sizes may produce different empirical results. However, how to choose an ideal window size is still an open question. We investigate how the window size affects the predictive performance of volatility. The empirical results show that the loss function for volatility prediction takes on a U-shape as the window size increases. This suggests that if the window size is chosen too large or too small, the loss function tends to be large and the model's predictive accuracy decreases. A window size of between 1000 and 2000 observations is ideal for various assets because it can produce relatively minimal forecast errors. From an asset allocation perspective, a mean–variance investor can obtain sizeable utility by using a model with a low loss function value for her portfolio. Moreover, the results are robust in a variety of settings.

Jel Classification: C22, C58, G17s

1 | Introduction

Stock market volatility is critical to asset pricing, asset allocation, derivative pricing, hedging, and risk management (see, e.g., Andersen, Bollerslev, and Diebold 2007; Bollerslev et al. 2018). Yet actual volatility is unobservable and therefore requires a proxy measure. With the abundance and availability of intraday data, Andersen and Bollerslev (1998) propose realized volatility/variance (hereafter RV) as a proxy for the integrated variance, which can be measured as the sum of squared intraday returns. And RV is robust to market microstructure effects from both theoretical and empirical perspectives. The main advantage of RV is that it is directly observable, allowing researchers to measure it and understand its dynamics. As a result, a large number of studies have investigated the prediction of stock market volatility (see, e.g., Ghysels, Santa-Clara, and Valkanov 2006; Andersen, Bollerslev, and Diebold 2007; Corsi 2009; Patton and Sheppard 2015; Ma et al. 2017; Wei et al. 2017; Zhang, Wei, and Liu 2019; He et al. 2021; Jiang, Tang, and Liu 2023).

It is worth noting that when predicting daily RV, the rolling window is usually selected. The rolling window always contains a specific number of samples and allows parameters to change over time. However, there are no broadly accepted guidelines for how to select the window size (Welch and Goyal 2008). The window size has an impact on how well the rolling window forecast (Rossi and Inoue 2012). The choice of window size requires a balance between using too little data, which may lead to inaccurate estimation, and too much data, which may leave too few observations during prediction. If the window size is too long, it may not be able to show the structural breaks in volatility. This can cause the effects before and after the structural breaks to cancel each other out. And because there are fewer observations in the evaluation window, this comes at the cost of lower power. If the window is too short, the amount of data used to estimate the parameters is too small, and the parameters obtained may be inaccurate. Motivated by these, we perform out-of-sample prediction using rolling windows with various window sizes to determine the optimal choice of window size.

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The study reveals several findings. The loss functions for out-of-sample predictions first decrease and then stabilize as the window size increases while predicting the volatility of the US stock market. The optimal window size for the model should be around 1200 observations. This is because it produces smaller loss functions and permits the use of additional data for out-of-sample forecasting. Second, as the window size increases, the loss functions for out-of-sample prediction in the Chinese stock market and the WTI market follow a U-shape, decreasing and then increasing. Moreover, when the model's loss function is relatively small, the model passes the Diebold and Mariano (1995) (hereafter DM) test at the 10% significance level. Third, applying the model with lower loss functions for portfolio analysis can result in higher utility gains for mean–variance investors. Finally, the results are robust to different evaluation periods, alternative volatility estimators, and a model with jump components.

Our main contributions are as follows: First, rolling windows are typically preferred by the finance literature in the body of existing literature for volatility forecasting (see, e.g., Bollerslev, Patton, and Quaedvlieg 2016; Degiannakis and Filis 2017; Ma, Wahab, and Zhang 2019; Zhang, Ma, and Liao 2020; He et al. 2021). However, the choice of the window size is arbitrary. We show that as the window size increases, the loss function for volatility prediction takes on a U-shape. The DM test is not significant, and the loss functions typically have large values when the window size is too large or too small.

The second and more important contribution is that in this study, we find the optimal window size interval that optimizes the model's forecasting performance. For various assets, a window size of between 1000 and 2000 observations is ideal because it can produce relatively minimal forecast errors. Specifically, selecting a rolling window size of roughly 1200 observations minimizes the model's loss in predicting the volatility of the US stock market. In forecasting the volatility of the Chinese stock market and the WTI oil market, the window sizes should be set to roughly 1500 to 2000 observations and roughly 1500 observations, respectively.

Our paper closes to the related literature on the choice of window size. Pesaran and Timmermann (2007) propose cross-validation and forecast combination methods to deal with uncertainty in the size of the estimation window. They use the cross-validation approach to deal with the selection of a single estimation window. For multiple breaks especially when the breakpoints are too small to be detected, they find that the combination and pooling methods work well. Rossi and Inoue (2012) aim to introduce a novel method for assessing the predictive performance of models that is robust to estimate window size selection. Hansen and Timmermann (2012) focus on the p-values of the DM test statistic estimated via a recursive window. They investigate how to choose the sample split point that defines the beginning of the assessment period when presenting out-of-sample performance. Inoue, Jin, and Rossi (2017) propose a new method for choosing the size of the rolling window for predicting models with potential breaks. To deal with parameter instability, they specify the parameters as smooth functions of time, whereas the functional form is unknown. Wu (2021) investigates the choice of rolling window size for cryptocurrencies in the face of structural breaks using a Markov-switching GARCH model. Complementary to their study, we directly investigate exactly which window size leads to optimal prediction performance. Similar to cross-validation in statistics, we consider the window size as a parameter and then find one that makes the model perform optimally.

The remainder of the paper is organized as follows. Section 2 introduces the forecasting methodology and the sources of data. Section 3 presents the empirical results. Section 4 considers a series of robustness checks. Finally, Section 5 concludes.

2 | Methodology

2.1 | RV Measure

With the availability of high-frequency data, we can use the sum of squared intraday stock returns to measure the daily RV, as proposed by Andersen and Bollerslev (1998). Statistically, we can express the RV on trading day t as follows:

$$RV_t = \sum_{i=1}^{N} r_{i,t}^2, \tag{1}$$

where $r_{i,t}$ is the ith intraday stock market return for day t. $N = 1/\Delta$, Δ is the sampling frequency.

2.2 | Heterogeneous autoregressive model for realized volatility (HAR-RV) Model

The HAR-RV, proposed by Corsi (2009), is the most prevalent benchmark volatility model. This model has some ideal properties. First, it takes into account some stylized facts about asset return volatility, such as long memory and multiscaling behavior. Additionally, it is simple to implement because there are only three predictors, including daily, weekly, and monthly RVs. Specifically, the model can be shown as follows:

$$RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w RV_{t-5:t-1} + \beta_m RV_{t-22:t-1} + \varepsilon_t, \tag{2}$$

where $RV_{t-h:t-1} = \left(\frac{1}{h}\right)\left(RV_{t-h} + \cdots + RV_{t-1}\right)$. Particularly, RV_{t-1}, $RV_{t-5:t-1}$, and $RV_{t-22:t-1}$ denote the daily, weekly, and monthly RVs, respectively. ε_t is a disturbance term.

We divide the whole sample into two parts: the in-sample period (estimation period) and the out-of-sample period (evaluation period) to statistically test the model's predictive performance. We focus on the predictive performance of three models: the HAR-RV-bench 1 model, the HAR-RV-bench 2 model, and the HAR-RV-size model. The first two models serve as competing models in our analysis. We suppose that there are p out-of-sample predictions during the evaluation period. The first forecast follows an estimation from a sample $T-q-m+1, \ldots, T-q$, and the last forecast follows an estimation from a sample $T-R+1, \ldots, R+q-1$, and $T=R+q$, where T is the sample length. Specifically, the HAR-RV-bench 1

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model's parameters are estimated using observations dated t, ..., $t+R-1$, for $t=1$, ..., q, where $m=T-q$. The HAR-RV-bench 2 model's parameters are based on a rolling window of fixed size, where the fixed size is 500 observations, namely $m=500$. In the HAR-RV-size model, the model's parameters are estimated using samples of m observations, dated $T-q-m+1$, ..., $T-q$, where the values of m range from 500 to $T-q$.

2.3 | Diebold–Mariano Test

The purpose of this paper is to compare the performance of predicting volatility generated by various window sizes. Therefore, we need to directly compare the out-of-sample predictive performance of the HAR-RV-size models with the corresponding competing models. In this sense, a one-to-one comparison method is very suitable for this study. In response to this motivation, we adopt a prevailing paired comparison method called the DM test, which is proposed by Diebold and Mariano (1995) and has been extensively used in numerous papers (see, e.g., Patton and Sheppard 2015; Buncic and Gisler 2017; Gong and Lin 2018; Zhang, Ma, and Wei 2019; Liang, Wei, and Zhang 2020).

We initially describe three loss functions: quasi-likelihood (QLIKE), mean square error (MSE), and mean absolute error (MAE), before performing the DM test. Their statistical expressions are as follows:

$$QLIKE: L\left(\widehat{RV}_t, RV_t\right) = log\left(\widehat{RV}_t\right) + \frac{RV_t}{\widehat{RV}_t}, \tag{3}$$

$$MSE: L\left(\widehat{RV}_t, RV_t\right) = \left(\widehat{RV}_t - RV_t\right)^2, \tag{4}$$

$$MAE: L\left(\widehat{RV}_t, RV_t\right) = \left|\widehat{RV}_t - RV_t\right|, \tag{5}$$

respectively, where RV_t is the actual RV on day t and $\widehat{RV}_t$ is the RV forecast generated by HAR-RV-size models or one of two competing models. Patton (2011) demonstrates that both QLIKE and MSE can withstand noise in the volatility proxy.

By now, we can calculate the DM statistic as

$$DM \text{ statistic} = \frac{\overline{d}}{\sqrt{Var(d)}}, \tag{6}$$

where $\overline{d} = \frac{1}{q}\sum_{t-m+1}^{m+q} d_t, d_t = L\left(\widehat{RV}_{b,t} - RV_t\right) - L\left(\widehat{RV}_{s,t} - RV_t\right)$; $Var(d)$ is the variance of $\{d_t\}_{t=m+1}^{m+q}$; $L\left(\widehat{RV}_{b,t} - RV_t\right)$ and $L\left(\widehat{RV}_{s,t} - RV_t\right)$ are the loss functions for the competing models and the HAR-RV-size model, respectively; and m and q are the length of the estimation period and the evaluation period.

The HAR-RV-size model performs better than competing models when the DM statistic is positive, and vice versa, following the definition of the DM statistic. This is because the HAR-RV-size model's forecast error is smaller under a particular loss function. Moreover, because the DM statistic is asymptotically standard normal, it is simple to determine the significance threshold for the DM test.

3 | Empirical Results

3.1 | Data

Martens (2001) discovers that a higher sampling frequency provides a more accurate daily volatility estimate. The effectiveness of high-frequency data can, however, be distorted by microstructure effects if the sampling frequency is too high. Therefore, in this paper, we choose a 5-min interval as our sampling frequency. This is because a 5-min sampling interval is widely used in a large number of studies (see, e.g., Andersen, Bollerslev, and Diebold 2007; Corsi, Pirino, and Renò 2010; Haugom et al. 2014; Wang et al. 2016; Ma et al. 2018; Zhang et al. 2019; He et al. 2022). Moreover, Liu, Patton, and Sheppard (2015) demonstrate that the 5-min RV cannot be surpassed by any other measure based on 400 volatility estimators covering 31 different financial assets across five asset classes.

We obtain the 5-min RV of three markets from the Oxford-Man Institute's Quantitative Finance Realized Library[1]: the China Shanghai Composite Index (SSEC), the US S&P 500 Index, and the West Texas Intermediate (WTI) oil markets. The sample period for the Chinese market is from February 16, 2000, to April 29, 2022; for the US market, it is from February 2, 2000, to May 2, 2022; and for the WTI oil market, it is from January 31, 2002, to February 27, 2022. There are outliers in the WTI oil market because of negative oil prices in 2020. We have performed a 99% winsorization on the WTI raw volatility data. This is an efficient way to deal with the impacts of outliers and extremes. Because of the difficulty of measuring outliers, they typically have substantial measurement errors (Bollerslev, Patton, and Quaedvlieg 2016).

Figure 1 plots the volatility of the three markets during the sample period. Affected by the global financial crisis in 2008 and China's stock market crash in 2015, the SSEC's RV is relatively more volatile in these two periods. In the US stock market, RV is high during the 2008 financial crisis. The RV of WTI crude oil is highly influenced by the Gulf War II in 2003, the global financial crisis in 2008, and the US–Iran conflicts in 2020.

3.2 | Out-Of-Sample Results

Figures 2, 3, and 4 show the loss function plots for out-of-sample volatility forecasts in the SSEC, S&P500, and WTI markets for various window sizes, respectively. The horizontal axis shows the number of observations used to estimate parameters, that is, the window size. The vertical axis represents the loss functions. Intuitively, a lower loss function means that the corresponding prediction is more accurate. For the out-of-sample evaluation period, we generate the last 2000 forecasts. Then, the window sizes for the HAR-RV-bench 1 and HAR-RV-bench 2 models are $T-q$ and 500 observations,

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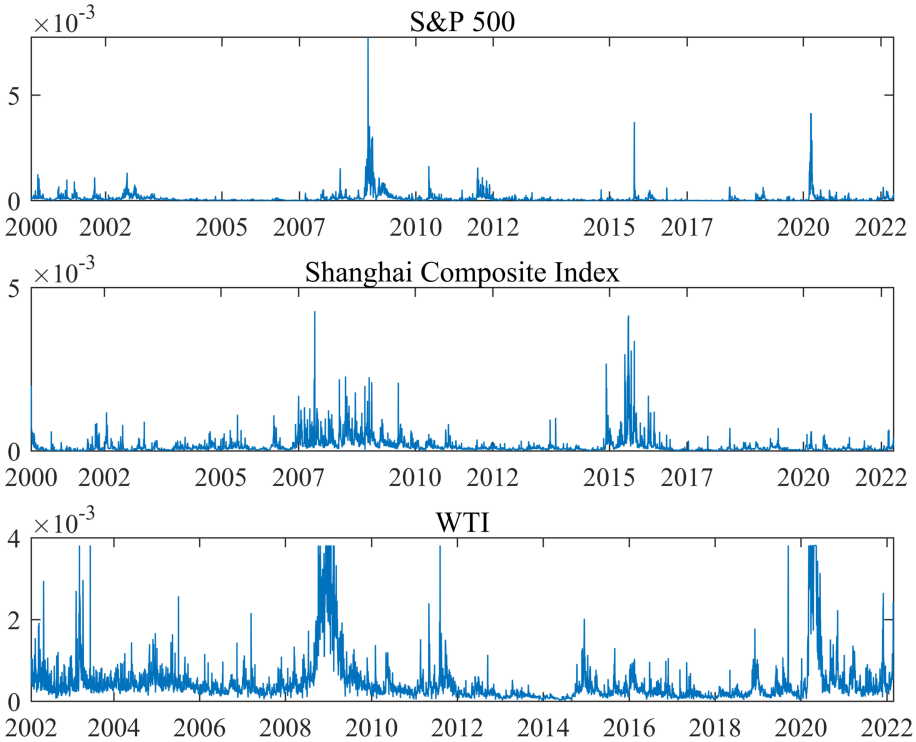

FIGURE 1 | The realized volatility of different markets. Time series of volatilities for the S&P 500, Shanghai Composite index, and WTI oil markets. The sample period for the S&P 500 is from February 2, 2000, to May 2, 2022. The sample period for the Shanghai Composite Index is from February 16, 2000, to April 29, 2022. The sample period for the WTI oil market is from January 31, 2002, to February 27, 2022.

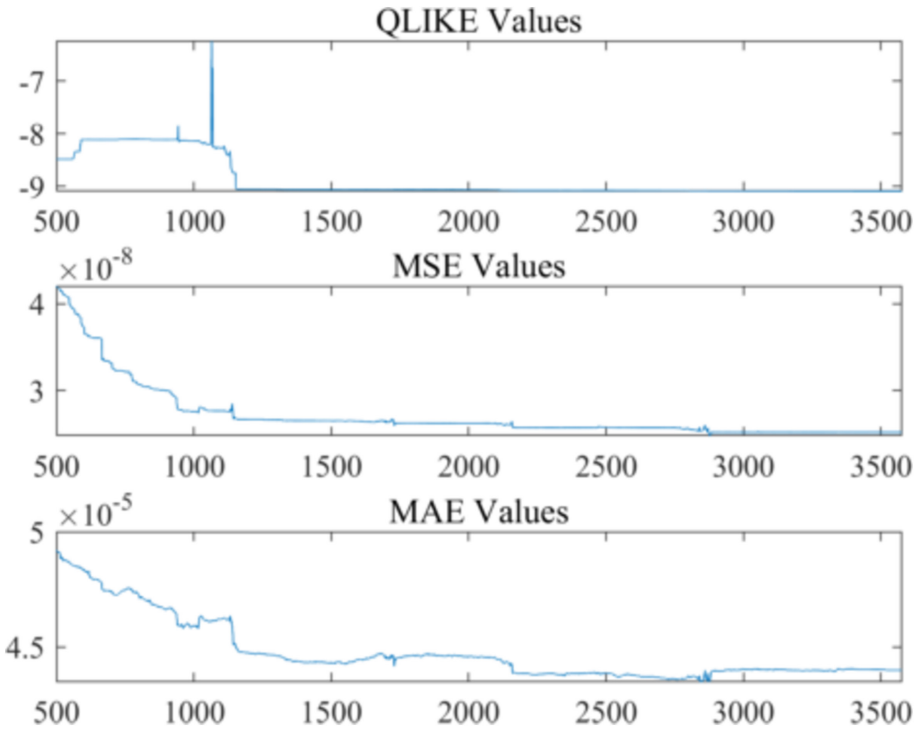

FIGURE 2 | Loss function value with the S&P 500 Index. The graph plots the loss functions obtained from out-of-sample forecasting of the S&P 500 Index's volatility. The horizontal axis shows the number of observations used to estimate parameters, and the vertical axis represents the loss functions. The whole sample period, containing 5576 observations, spans from February 2, 2000, to May 2, 2022, whereas the out-of-sample evaluation period consists of the last 2000 observations.

respectively, and the window sizes for the HAR-RV-size model vary continuously from 500 to $T-q$ (where the evaluation period q is 2000).

As window sizes increase, loss functions for volatility prediction in the US stock market gradually decrease and stabilize, as shown in Figure 2. To test whether the performance of the

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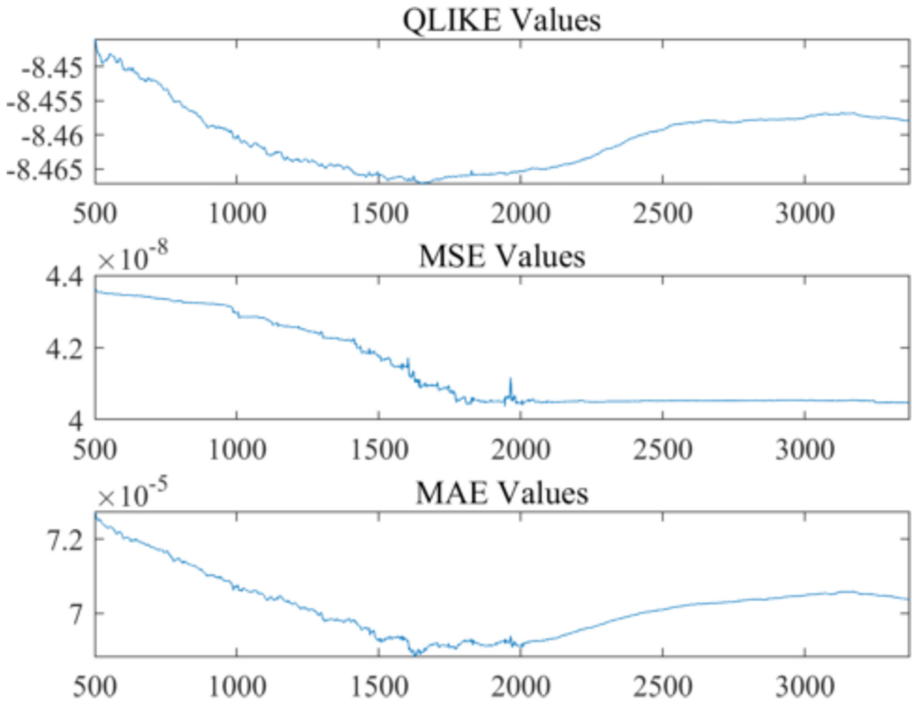

FIGURE 3 | Loss function value with the Shanghai Composite Index. The graph plots the loss functions obtained from out-of-sample forecasting of the Shanghai Composite Index's volatility. The horizontal axis shows the number of observations used to estimate parameters, and the vertical axis represents the loss functions. The whole sample period, containing 5369 observations, spans from February 16, 2000, to April 29, 2022, whereas the out-of-sample evaluation period consists of the last 2000 observations.

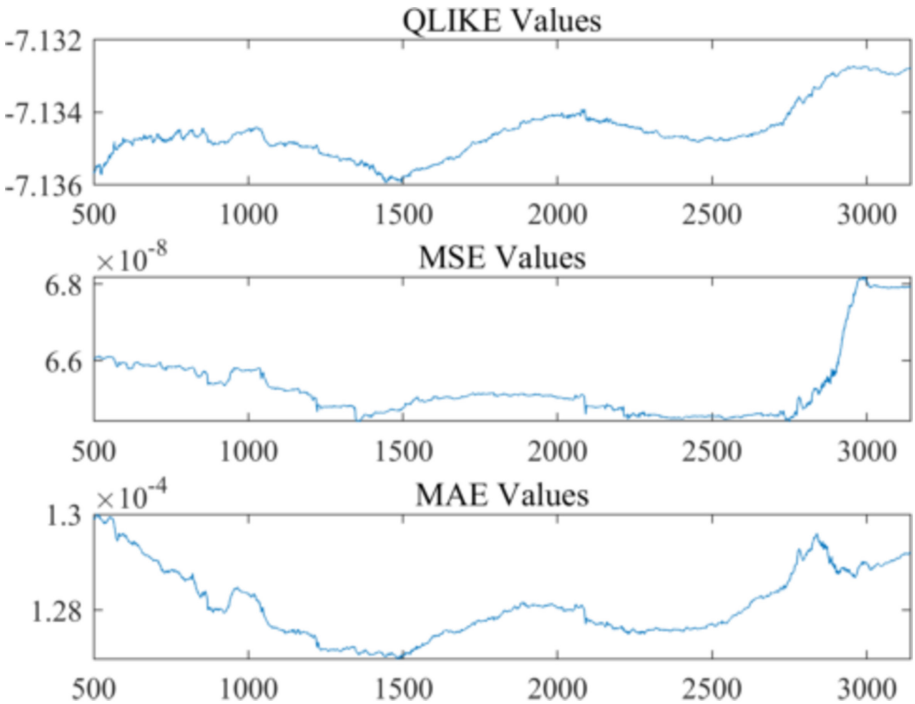

FIGURE 4 | Loss function value with WTI. The graph plots the loss functions obtained from out-of-sample forecasting of the volatility of the WTI oil market. The horizontal axis shows the number of observations used to estimate parameters, and the vertical axis represents the loss functions. The whole sample period, containing 5143 observations, spans from January 31, 2002, to February 27, 2022, whereas the out-of-sample evaluation period consists of the last 2000 observations.

optimal window size is statistically significant compared with the HAR-RV-bench 1 and HAR-RV-bench 2 models, we perform a DM test. Table 1 reports the results of the DM test for the US stock market.[2] Taking the QLIKE loss function as an example and using the HAR-RV-bench 1 model as a competitive model for the DM test, when the window size is 1153 observations,

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TABLE 1 | DM test results based on S&P 500 index.

	QLIKE	MSE	MAE	QLIKE	MSE	MAE
Size	**Panel A: HAR-RV-bench 1 model**			**Panel B: HAR-RV-bench 2 model**		
500	−2.155	−1.199	−1.858			
600	−2.128	−1.205	−1.834	−2.081	1.166	1.772**
700	−2.126	−1.285	−1.716	−2.076	1.082	1.805**
800	−2.137	−1.323	−1.866	−2.104	1.058	1.399**
900	−2.124	−1.297	−1.597	−2.07	1.056	1.644**
1000	−2.097	−0.833	−1.324	−1.97	1.041	1.297**
1100	−1.978	−0.91	−1.626	−1.342	1.097	1.453**
1200	−6.774	−1.177	−0.905	2.063**	1.15	1.898**
1300	−6.567	−1.152	−0.775	2.069**	1.15	1.93**
1400	−5.896	−1.101	−0.415	2.083**	1.156	2.047**
1500	−5.522	−1.093	−0.405	2.092**	1.155	2.044**
1600	−5.429	−1.095	−0.565	2.095**	1.153	1.976**
1700	−5.133	−1.09	−0.696	2.1**	1.158	1.958**
1800	−6.337	−0.976	−0.853	2.089**	1.134	1.753**
1900	−6.162	−0.959	−0.854	2.093**	1.136	1.755**
2000	−6.099	−0.964	−0.811	2.096**	1.138	1.777**
2100	−6.312	−1.063	−0.704	2.103**	1.139	1.8**
2200	−7.153	−0.882	0.276	2.114**	1.167	1.972**
2300	−6.887	−0.853	0.286	2.118**	1.168	1.971**
2400	−7.405	−0.869	0.302	2.118**	1.168	1.973**
2500	−7.967	−1.03	0.342	2.117**	1.163	1.956**
2600	−7.546	−1.005	0.489	2.12**	1.164	1.98**
2700	−6.487	−0.98	0.741	2.128**	1.166	2.021**
2800	−6.106	−1.183	0.713	2.134**	1.169	1.986**
2900	−6.113	0.575	0.398	2.138**	1.201	1.864**
3000	−5.799	0.391	−0.583	2.141**	1.2	1.84**
3100	−5.208	0.296	−0.327	2.145**	1.199	1.846**
3200	−3.873	0.961	0.352	2.148**	1.199	1.858**
3300	−3.234	1.115	0.95	2.151**	1.2	1.87**
3400	−3.559	−0.128	−3.923	2.153**	1.199	1.843**
3500	−1.919	0.433	−1.72	2.154**	1.2	1.855**

Note: This table reports the out-of-sample evaluation results by the Diebold–Mariano (DM) test for the HAR-RV-size models with variable window sizes relative to the competing models with fixed window sizes of 3576 and 500 in the US stock market. The first column represents the window size corresponding to the model. The DM statistic and its *p*-value are reported based on three loss functions: QLIKE, MSE, and MAE. A positive DM statistic indicates the HAR-RV-size model exhibits better out-of-sample forecasting performance, and vice versa. The whole sample period, containing 5576 observations, spans from February 2, 2000, to May 2, 2022, whereas the out-of-sample evaluation period consists of the last 2000 observations.
*, **, and ***denote rejections of the null hypothesis at 10%, 5%, and 1% significance levels, respectively.

the loss function is −9.05. The decrease in the loss function is extremely small with the increase in the window size. As a result, the increase in the window size has almost no effect on improving the performance of the model. Hence, even if the HAR-RV-size model fails the DM test when the HAR-RV-bench 1 model is chosen as the benchmark model, we can find that choosing a shorter window size can retain more observations for out-of-sample evaluation with little change in the model loss.

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When the HAR-RV-bench 2 model is used as a competing model for the DM test, the DM test is significant at the 5% level when the window size is greater than 1153 observations. This result suggests that the window size should be chosen around 1200 observations and not be too small (large loss functions) or too large (insignificant changes in loss functions) for volatility forecasting in the US stock market.

One notable observation from Figures 3 and 4 is the prevalence of a U-shaped pattern in the loss function for volatility prediction, wherein the loss function decreases initially and then increases as the window size increases. It indicates that there is an optimal window size that minimizes the loss of the model for volatility forecasting in the Chinese stock market and the WTI oil market.

TABLE 2 | DM test results based on the China Shanghai Composite Index.

	QLIKE	MSE	MAE	QLIKE	MSE	MAE
Size	**Panel A: HAR-RV-bench 1 model**			**Panel B: HAR-RV-bench 2 model**		
500	−3.867	−1.719	−2.157			
600	−3.034	−1.644	−1.551	2.928***	2.687***	5.52***
700	−2.394	−1.621	−1.33	2.989***	2.073**	4.895***
800	−1.267	−1.594	−1.044	4.12***	1.995**	5.408***
900	0.43	−1.571	−0.623	5.488***	1.92**	6.499***
1000	1.264	−1.474	−0.36	5.803***	1.74**	5.748***
1100	2.42***	−1.409	−0.044	6.251***	1.725**	5.985***
1200	3.525***	−1.318	0.138	6.405***	2.161**	6.187***
1300	4.043***	−1.254	0.5	6.4***	1.705**	5.318***
1400	4.675***	−1.172	0.761	6.55***	1.9**	5.254***
1500	5.489***	−1.065	1.708**	6.734***	1.915**	4.969***
1600	6.356***	−0.944	1.833**	6.951***	2.09**	4.93***
1700	6.688***	−0.514	2.572***	6.796***	2.253**	4.648***
1800	6.458***	−0.226	2.709***	6.692***	2.311**	4.229***
1900	6.624***	−0.029	3.546***	6.941***	2.273**	3.998***
2000	6.85***	0.094	6.668***	6.846***	1.782**	3.464***
2100	6.763***	−0.295	7.748***	6.847***	1.701**	3.194***
2200	6.01***	−0.337	6.733***	6.667***	1.702**	2.989***
2300	4.108***	−0.545	5.173***	6.217***	1.692**	2.768***
2400	2.424***	−0.877	3.602***	5.6***	1.681**	2.572***
2500	1.343*	−0.937	2.529***	5.053***	1.671**	2.429***
2600	0.241	−1.068	1.364*	4.482***	1.664**	2.291**
2700	0.119	−1.037	1.029	4.322***	1.669**	2.242**
2800	−0.157	−1.198	0.188	4.323***	1.669**	2.242**
2900	−0.375	−1.303	−0.488	4.174***	1.662**	2.165**
3000	−0.956	−1.418	−1.728	4.088***	1.659**	2.116**
3100	−2.577	−1.534	−2.777	3.949***	1.664**	2.051**
3200	−3.664	−1.255	−4.12	3.726***	1.675**	2.02**
3300	−4.151	−0.135	−4.831	3.676***	1.668**	2.016**

Note: This table reports the out-of-sample evaluation results by the Diebold–Mariano (DM) test for the HAR-RV-size models with variable window sizes relative to the competing models with fixed window sizes of 3369 and 500 in the Chinese stock market. The first column represents the window size corresponding to the model. The DM statistic and its *p*-value are reported based on three loss functions: QLIKE, MSE, and MAE. A positive DM statistic indicates the HAR-RV-size model exhibits better out-of-sample forecasting performance, and vice versa. The whole sample period, containing 5369 observations, spans from February 16, 2000, to April 29, 2022, whereas the out-of-sample evaluation period consists of the last 2000 observations.
*, **, and ***denote rejections of the null hypothesis at 10%, 5%, and 1% significance levels, respectively.

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TABLE 3 | DM tests based on the WTI oil market.

	QLIKE	MSE	MAE	QLIKE	MSE	MAE
Size	**Panel A: HAR-RV-bench 1 model**			**Panel B: HAR-RV-bench 2 model**		
500	1.917**	0.818	−0.424			
600	1.478*	0.901	−0.22	15.808***	0.686	20.603***
700	1.431*	0.957	0.098	16.983***	0.364	20.667***
800	1.42*	1.029	0.379	17.161***	0.693	20.902***
900	1.643*	1.224	0.805	17.346***	0.874	20.947***
1000	1.362*	1.07	0.601	17.44***	0.826	20.721***
1100	1.75**	1.334	1.155	14.229***	0.999	20.414***
1200	1.909**	1.396	1.23	12.862***	0.925	20.196***
1300	2.258**	1.543	1.452*	12.843***	0.876	19.706***
1400	2.5***	1.589	1.491*	9.678***	0.963	18.857***
1500	2.716***	1.514	1.528*	9.706***	0.982	18.839***
1600	2.54***	1.398	1.293*	9.688***	1.022	18.674***
1700	2.288**	1.41	1.218	9.705***	1.042	18.628***
1800	1.849**	1.394	1.038	9.733***	1.038	18.524***
1900	1.618*	1.414	0.87	9.778***	1.05	18.492***
2000	1.458*	1.473	0.925	9.857***	1.044	18.324***
2100	1.675**	1.576	1.12	9.867***	1.044	18.175***
2200	2.105**	1.629	1.258	9.376***	1.049	17.868***
2300	2.509***	1.642	1.283	7.892***	1.048	16.567***
2400	2.816***	1.698**	1.299	7.859***	1.027	16.385***
2500	3.093***	1.681**	1.197	7.916***	1.024	16.315***
2600	3.222***	1.69**	0.988	7.718***	1.03	16.112***
2700	3.215***	1.729**	0.728	7.697***	1.037	15.911***
2800	2.316**	1.727**	0.084	7.724***	1.111	15.614***
2900	1.059	2.399***	0.562	6.762***	1.693**	11.526***
3000	−0.352	−1.002	1.318*	−0.9	2.005**	9.497***
3100	2.742***	0.62	2.177**	−0.9	2.005**	9.405***

Note: This table reports the out-of-sample evaluation results by the Diebold–Mariano (DM) test for the HAR-RV-size models with variable window sizes relative to the competing models with fixed window sizes of 3143 and 500 in the WTI oil market. The first column represents the window size corresponding to the model. The DM statistic and its *p*-value are reported based on three loss functions: QLIKE, MSE, and MAE. A positive DM statistic indicates the HAR-RV-size model exhibits better out-of-sample forecasting performance, and vice versa. The whole sample period, containing 5143 observations, spans from January 31, 2002, to February 27, 2022, whereas the out-of-sample evaluation period consists of the last 2000 observations.
*, **, and ***denote rejections of the null hypothesis at 10%, 5%, and 1% significance levels, respectively.

Tables 2 and 3 report the out-of-sample results of the DM test for the Chinese stock market and the WTI oil market, respectively. In the Chinese stock market, when the HAR-RV-bench 1 model is used as the competing model and the loss functions are QLIKE and MAE, the DM tests of the HAR-RV-size model are significant at the 10% level for window sizes ranging from 1007 to 2503 observations and 1484 to 2620 observations, respectively. When the HAR-RV-bench 2 model is the competing model, the DM tests of the HAR-RV-size model are almost always significant at the 5% level, regardless of which loss function is taken. Combining the results of Figure 3 and Table 2, when forecasting the volatility of the Chinese stock market, the window size should be chosen about 2000 observations.

In the WTI market, when using the HAR-RV-bench 1 model as the competing model and the loss function is QLIKE, the DM tests of the HAR-RV-size model are significant at the 10% level with a window size ranging from 500 to 2859 observations. The DM tests with MSE and MAE as loss functions are significant at the 10% level, with window sizes ranging from

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1064 to 2956 observations and from 1220 to 1618 observations, respectively. When using the HAR-RV-bench 2 model as the competing model, the DM tests with QLIKE and MAE as loss functions are almost always significant at the 5% level, and the DM tests with MSE as a loss function are significant at the 10% level when the window size is larger than 2866 observations. Combining the results of Figure 4 and Table 3, when forecasting the volatility of the WTI oil market, the window size should be chosen around 1500 observations.

In general, a window size of 1000 to 2000 observations is ideal for minimizing forecast errors across different assets. When the window size is too small, the inaccurate estimation of the model's parameters leads to a larger model loss. If the window size is too large, the structural breaks in the data may not be visible. This can cause the effects before and after the breaks to cancel out, resulting in a less accurate model prediction.

3.3 | Portfolio Performance

We further investigate the economic value of RV forecasts generated by predictive models in an asset allocation exercise because investors are more concerned about whether it can produce economic gains. Many approaches rely on return and volatility forecasts (see, e.g., Fleming, Kirby, and Ostdiek 2001, 2003; Campbell and Thompson 2008; Rapach, Strauss, and Zhou 2010; Wang et al. 2016; Zhang et al. 2019; He et al. 2021; Zhang and Wang 2023). However, Bollerslev et al. (2018) propose a portfolio exercise that relies exclusively on forecasting volatility. This is attractive because predicting returns is notoriously difficult (see, e.g., Campbell and Thompson 2008; Welch and Goyal 2008; Rapach, Strauss, and Zhou 2010). Specifically, following Bollerslev et al. (2018), we assume that a mean–variance investor will allocate assets between risky assets (i.e., stocks) and risk-free assets (i.e., risk-free bills) with a fixed Sharpe ratio (SR hereafter).

In the portfolio exercise, a mean–variance investor will allocate a portion w_t of her current portfolio to stocks with an expected return of r_{t+1}, and the remainder to risk-free bills with a return of r_t^f. As a result, her portfolio will earn the following returns for her:

$$r_{t+1}^p = w_t r_{t+1} + (1 - w_t) r_t^f = w_t r_{t+1}^e + r_t^f, \tag{7}$$

where $r_{t+1}^e = r_{t+1} - r_t^f$.

The expression of the expected utility is as follows:

$$U(w_t) = w_t E_t(r_{t+1}^e) - \frac{\gamma}{2} w_t^2 Var(r_{t+1}^e), \tag{8}$$

where γ is the mean–variance investor's risk aversion coefficient and $Var(r_{t+1}^e) = E_t(RV_{t+1})$. For volatility forecasting, Bollerslev et al. (2018) propose the conditional SR, which can be measured as $SR = \frac{E_t(r_{t+1}^e)}{\sqrt{E_t(RV_{t+1})}}$ and be constant. Thus, the expected utility can be expressed as:

$$U(w_t) = w_t SR \sqrt{E_t(RV_{t+1})} - \frac{\gamma}{2} w_t^2 E_t(RV_{t+1}). \tag{9}$$

To fulfill the goal of maximum expected utility, the investor will distribute the stock weight to

$$w_t^* = \frac{SR/\gamma}{\sqrt{E_t(RV_{t+1})}}. \tag{10}$$

Simply expressed, the optimal target for investors is the volatility of $\frac{SR}{\gamma}$, where $\sqrt{Var(w_t^* r_{t+1}^e)} = \frac{SR}{\gamma}$ is the conditional standard deviation of the portfolio's risky component. When the forecasted volatility risk of $\sqrt{E_t(RV_{t+1})}$ is larger than the optimal target of $\frac{SR}{\gamma}$ (i.e., $w_t^* < 1$), then the investor invests only a portion of her assets in the risky asset of stocks. Conversely, when $\sqrt{E_t(RV_{t+1})}$ is less than $\frac{SR}{\gamma}$ (i.e., $w_t^* > 1$), the investor needs to use leverage to reach her risk target.

This case in turn leads to an expected utility of

$$U(w_t^*) = \frac{SR^2}{2\gamma}. \tag{11}$$

However, $\sqrt{E_t(RV_{t+1})}$ is not available in real life. By using the RV forecast $\widehat{RV}_{t+1}$ for day $t+1$, we obtain the expected utility as follows:

$$U\left(\widehat{RV}_{t+1}\right) = \frac{SR^2}{\gamma}\left(\frac{\sqrt{RV_{t+1}}}{\sqrt{\widehat{RV}_{t+1}}} - \frac{1}{2}\frac{RV_{t+1}}{\widehat{RV}_{t+1}}\right). \tag{12}$$

We estimate the average utility for the out-of-sample period, which is measured as follows:

$$\overline{U}\left(\widehat{RV}\right) = \frac{1}{q}\sum_{t=m}^{m+q-1}\frac{SR^2}{\gamma}\left(\frac{\sqrt{RV_{t+1}}}{\sqrt{\widehat{RV}_{t+1}}} - \frac{1}{2}\frac{RV_{t+1}}{\widehat{RV}_{t+1}}\right), \tag{13}$$

where m and q stand for the length of the in-sample period and out-of-sample period, respectively. According to Bollerslev et al. (2018), we take the risk aversion coefficient and the annualized Sharpe ratio as $\gamma = 2$ and $SR = 0.4$, respectively. Therefore, $U(w_t^*) = 4\%$. This result implies that the investor is willing to pay 4% of her assets to obtain the w_t^* portfolio of risky assets instead of investing exclusively in risk-free bills.

Table 4 reports the portfolio performance of the HAR-RV-size model in the three markets, as measured by the average realized utility.[3] At different window sizes, all volatility forecasts generate almost positive utility, implying that using volatility forecasts to allocate portfolios is more profitable than simply investing in risk-free bills. For the US market, when the window size is larger than 1200, the utility of the model hardly changes as the window size increases, which is consistent

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TABLE 4 | Portfolio performance.

Size	Utility	Size	Utility	Size	Utility
Panel A: S&P 500 index		**Panel B: SSEC index**		**Panel C: WTI oil market**	
500	1.3	500	3.587	500	3.794
600	−0.089	600	3.595	600	3.792
700	−0.094	700	3.598	700	3.791
800	−0.107	800	3.602	800	3.79
900	−0.098	900	3.608	900	3.791
1000	−0.002	1000	3.61	1000	3.790
1100	0.51	1100	3.614	1100	3.790
1200	3.401	1200	3.617	1200	3.790
1300	3.404	1300	3.617	1300	3.791
1400	3.412	1400	3.619	1400	3.791
1500	3.416	1500	3.621	1500	3.792
1600	3.417	1600	3.623	1600	3.792
1700	3.422	1700	3.624	1700	3.791
1800	3.415	1800	3.625	1800	3.790
1900	3.42	1900	3.625	1900	3.790
2000	3.421	2000	3.625	2000	3.789
2100	3.427	2100	3.625	2100	3.789
2200	3.441	2200	3.624	2200	3.790
2300	3.444	2300	3.622	2300	3.790
2400	3.444	2400	3.620	2400	3.791
2500	3.445	2500	3.618	2500	3.791
2600	3.447	2600	3.617	2600	3.791
2700	3.45	2700	3.617	2700	3.791
2800	3.453	2800	3.616	2800	3.79
2900	3.457	2900	3.616	2900	3.789
3000	3.459	3000	3.616	3000	3.789
3100	3.461	3100	3.615	3100	3.789
3200	3.462	3200	3.616	3143	3.789
3300	3.464	3300	3.616		
3400	3.465	3369	3.617		
3500	3.466				
3576	3.466				

Note: Panels A, B, and C of the table report portfolio performance for the S&P 500, the Shanghai Composite index, and the WTI crude oil market, respectively. The first column of each panel represents the window size corresponding to the model. We suppose that a mean–variance investor will allocate her assets between stock and risk-free bills through various RV forecasts, where the risk aversion coefficient is 2 and the constant Sharpe ratio is 0.4. The first and last rows of each panel present the utility of the two competing models (HAR-RV-bench1 and HAR-RV-bench2), respectively.

with the loss functions and DM test results shown in Table 1. More importantly, for the Chinese and WTI markets, the utility gains obtained by the HAR-RV-size model are larger than those obtained by the two competing models for window sizes where the model loss functions are low and the DM tests are significant. These results suggest that the HAR-RV-size model can help investors maximize economic gains in their portfolios when the model uses a window size with a low loss function.

3.4 | Discussion

The majority of scholars in the existing literature discuss the predictive performance of the model under different out-of-sample prediction windows, and the conclusions are usually robust. However, their robustness is based on comparisons between models or variables. That is, this robustness is relative to other (or benchmark) models or variables, and their proposed models or variables perform better under different window sizes. This paper is simply a comparison of model performance under different window sizes. For example, a paper proposes that model A performs best relative to model B and robustly demonstrates that model A performs best under different window sizes. However, it does not give the window size under which model A itself performs best.

This paper can answer this question. We demonstrate that the loss function for volatility prediction is U-shaped as the window size increases. This suggests that if the window size is chosen too large or too small, the loss function becomes larger and the model's forecasting accuracy decreases. A window size between 1000 and 2000 observations is ideal for various assets, as it produces relatively minimal forecast error.

4 | Robustness Analyses

4.1 | Alternative Evaluation Sizes

The forecasting window size is equally important in out-of-sample evaluation. Different window sizes may lead to completely different out-of-sample results. (see, e.g., Rossi and Inoue 2012; Inoue, Jin, and Rossi 2017). Therefore, we select another out-of-sample evaluation size to evaluate the robustness of the performance of the HAR-RV-size model. In particular, the out-of-sample evaluation period consists of the last 2500 observations.

Figure 5 reports the loss functions when using different estimation window sizes. We have the following two findings. First, the loss function plots for the three markets are similar to those when the evaluation period contains the last 2000 observations. As the window size increases, the loss functions of the US stock market exhibit a trend of gradually decreasing and then remaining stable. In general, the loss functions of the Chinese stock market and the WTI market roughly show a trend of decreasing and then increasing. Additionally, the window size intervals where the DM test is significant and the loss functions for the model are low are essentially identical.[4] Second, comparing the loss functions for different evaluation periods reveals that

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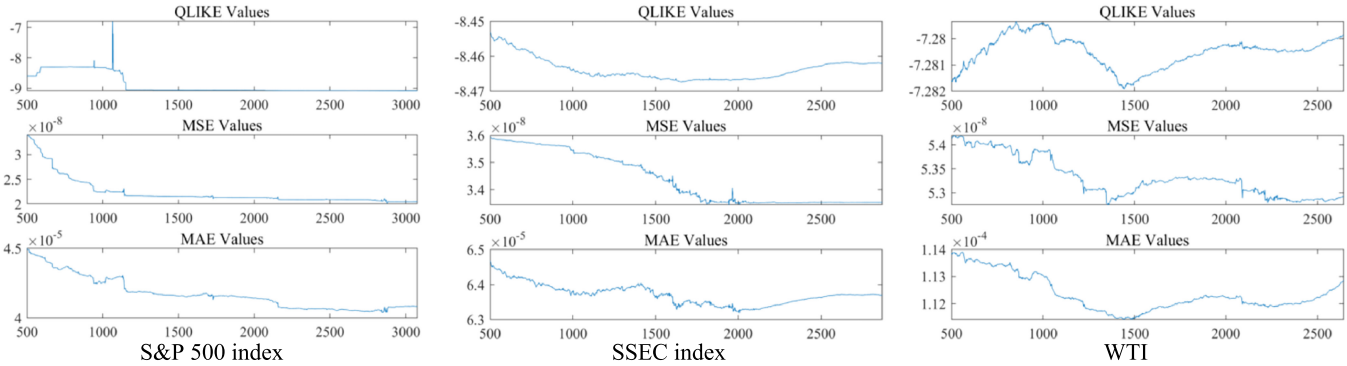

FIGURE 5 | Loss function value with alternative window size. The graph plots the loss functions for three markets. The horizontal axis shows the number of observations used to estimate parameters, and the vertical axis represents the loss functions. The left panel depicts a plot of the loss function derived from out-of-sample predictions of S&P 500 volatility. The middle and right panels show the loss function plot derived from out-of-sample predictions of the Shanghai Composite Index's volatility and the WTI oil volatility. The out-of-sample evaluation period consists of the last 2500 observations.

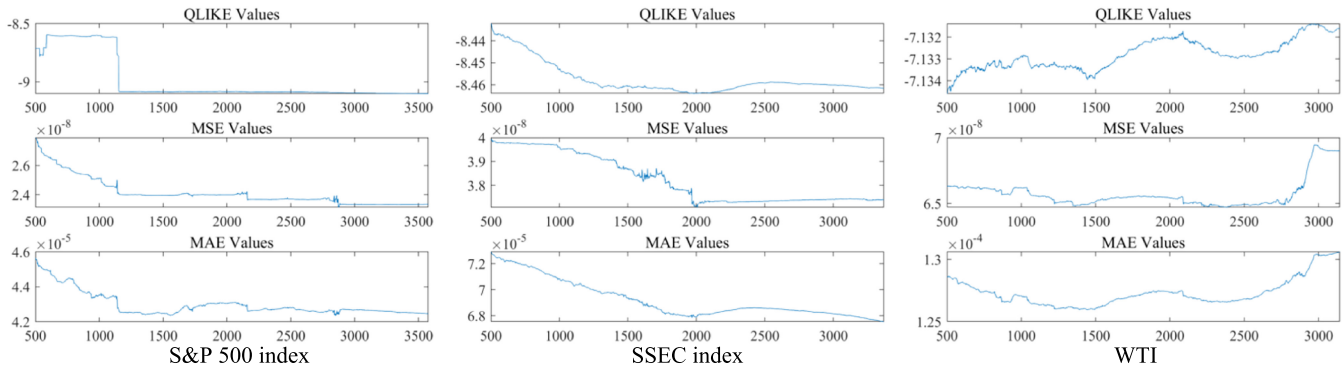

FIGURE 6 | Loss function value when the volatility estimator is RK. In this section, the realized kernel (RK), not the RV, is used as the volatility estimator. The horizontal axis shows the number of observations used to estimate parameters, and the vertical axis represents the loss functions. The left panel depicts a plot of the loss function derived from out-of-sample predictions of S&P 500 volatility. The middle and right panels show the loss function plot derived from out-of-sample predictions of the Shanghai Composite Index's volatility and the WTI oil volatility. The out-of-sample evaluation period consists of the last 2000 observations.

when the evaluation period contains the last 2500 observations, the loss functions of the model are smaller than those when the evaluation period contains the last 2000 observations. This implies that we ought to choose appropriate data for parameter estimation and, consequently, use more data for out-of-sample evaluation. The aforementioned findings demonstrate that our findings are robust to various evaluation sizes.

4.2 | Realized Kernel (RK)

Considering that actual volatility is unobservable and market microstructure noise impairs the measurement of RV (see, e.g., Bandi and Russell 2008; Andersen, Bollerslev, and Meddahi 2011; Liu, Patton, and Sheppard 2015), we consider the use of the RK proposed by Barndorff-Nielsen et al. (2008) to recheck the out-of-sample performance of the forecasting model. This volatility estimator is robust to market microstructure noise. RK is defined as follows:

$$RK_t = \sum_{h=-H}^{H} k\left(\frac{h}{H+1}\right)\gamma_h, \tag{14}$$

where

$$\gamma_h = \sum_{j=|h|+1}^{N} r_{t,j} r_{t,j-|h|}, \tag{15}$$

and $k(x)$ is the Parzen kernel function and can be provided by

$$k(x) = \begin{cases} 1-6x^2+6x^3 & 0 \le x < \frac{1}{2} \\ 2(1-x)^3 & \frac{1}{2} \le x \le 1 \\ 0 & x > 1 \end{cases}. \tag{16}$$

For a specific choice of H, please refer to Barndorff-Nielsen et al. (2009).

Figure 6 reports the results of loss functions when the volatility estimator is RK. Consistent with the above results, as the window size increases, the loss function of the US stock market shows a decrease and then tends to be stable. The loss function of the Chinese stock market and WTI oil market shows a decrease

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and then an increase. When the model generates volatility forecasts that have lower loss functions, their DM test is always significant at the 10% level. Therefore, the predictive power of the HAR-RV-size model is robust to alternative volatility estimators.

4.3 | HAR-RV-J Model

The importance of jumps is widely recognized in financial economics, and Corsi, Pirino, and Renò (2010) further find that jumps are crucial for forecasting volatility. Because of the availability of jump data, we construct the HAR-RV-J model using the jump measure proposed by Andersen, Bollerslev, and Diebold (2007), which is one of the mainstream models for forecasting volatility using jumps. The HAR-RV-J model is written as

$$RV_t = \varphi_0 + \beta_d RV_{t-1} + \beta_w RV_{t-5:t-1} + \beta_m RV_{t-22:t-1} + \beta_J J_{t-1} + \varepsilon_{t+1:t+h}, \tag{17}$$

where the jump component $J_t = max\{RV_t - BPV_t, 0\}$, $BPV_t = u_1^{-2} \sum_{i=2}^{M} |r_{t,i}||r_{t,i-1}|$ is the realized bi-power variation (BPV), and $u_1 = \sqrt{\frac{2}{\pi}}$.

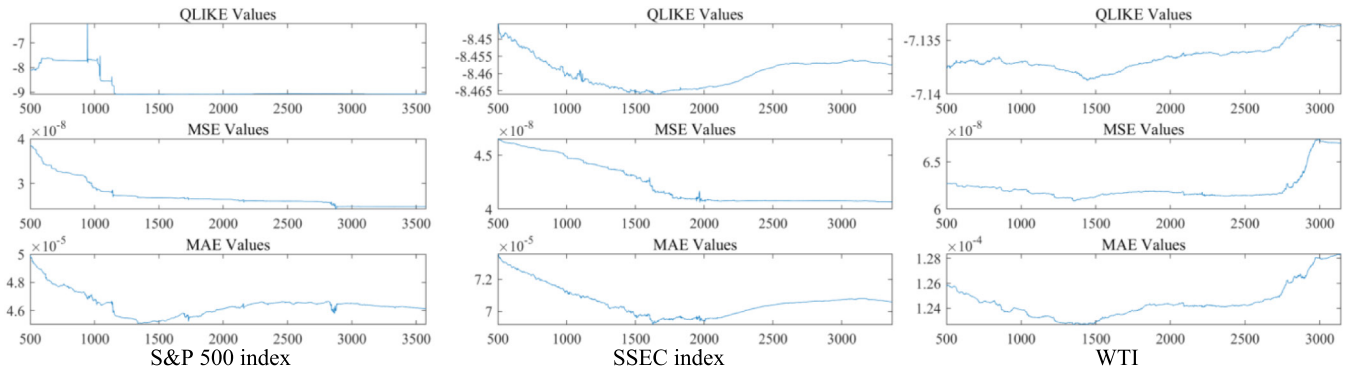

FIGURE 7 | Loss function value when the model is HAR-RV-J. The model used here is the HAR-RV-J model with the jump component. The horizontal axis shows the number of observations used to estimate parameters, and the vertical axis represents the loss functions. The left panel depicts a plot of the loss function derived from out-of-sample predictions of S&P 500 volatility. The middle and right panels show the loss function plot derived from out-of-sample predictions of the Shanghai Composite Index's volatility and the WTI oil volatility. The out-of-sample evaluation period consists of the last 2000 observations.

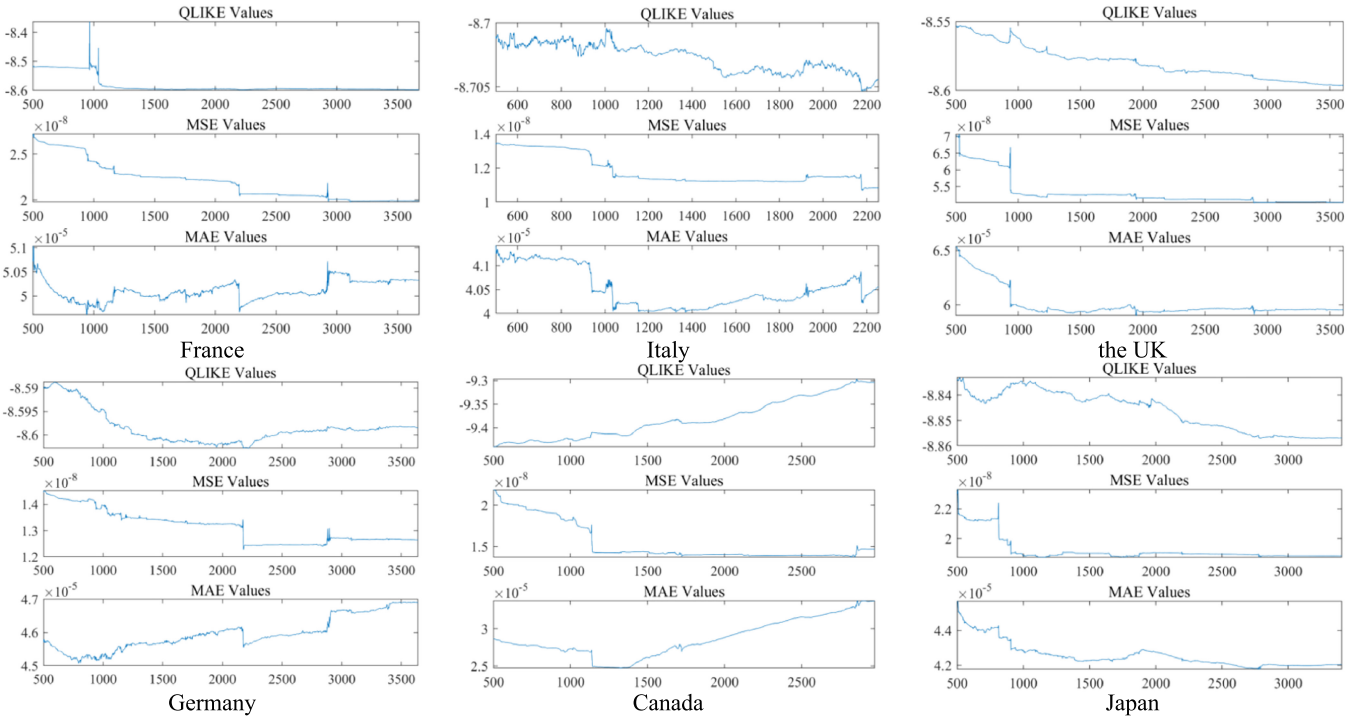

FIGURE 8 | Loss function of G7 countries. The graph plots the loss functions obtained from out-of-sample forecasting. The Group of Seven (G7) includes Canada, France, Germany, Italy, Japan, the United Kingdom, and the United States. The horizontal axis shows the number of observations used to estimate parameters, and the vertical axis represents the loss functions. The Italian sample began in June 2009, whereas the samples from other countries began in February 2000. All samples from countries are up to April 2022. The out-of-sample evaluation period encompasses the last 1000 observations for Italy and the last 2000 observations for the other five countries.

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Thus, we generate the forecasts by using a jump model. Figure 7 reports the loss functions of the HAR-RV-J-size model. Both the HAR-RV-J-type model and the HAR-RV-type model provide comparable loss function plots as the benchmark models. The HAR-RV-J-size model has lower loss functions than the two competing models, implying greater forecasting accuracy. This indicates that the HAR-RV-J-size model is superior to the HAR-RV-J-bench models. Therefore, our results are robust to the HAR-RV-J model.

4.4 | Alternative Markets

To demonstrate the robustness of our results, we extend the model to G7 country markets. The Group of Seven (G7) includes Canada, France, Germany, Italy, Japan, the UK, and the US. It is composed of the seven wealthiest and most advanced countries. Figure 8 shows the loss functions of the G7 countries. The Italian market is observed from June 1, 2009, to May 2, 2022, for a total of 3274 observations. The out-of-sample evaluation period in Italy consists of the last 1000 observations. The rest of the stock market starts in 2000, with more observations, and the out-of-sample analysis includes the last 2000 observations. We find that although the shape of the loss function is not U-shaped, the loss function is all low when the window length is 1000–2000 observations (except for the Japanese and UK stock markets, where the loss function is QLIKE). Therefore, a window size of between 1000 and 2000 observations is ideal for various assets. Our results are robust to different stock markets.

5 | Conclusion

We usually choose rolling windows for daily volatility forecasting. However, the choice of window size is an important and open problem. Therefore, in this paper, we perform volatility forecasting by using various window sizes. We mainly explore whether there exists an optimal window size that minimizes the loss of the volatility forecasting model. To this end, we build the HAR-RV-size model, plot the loss function of the models for various window sizes, and perform the DM test.

We find some interesting findings. First, as the window size increases, the loss functions for US stock market volatility forecasts gradually decrease and then stabilize. The loss function is lowest when the window size is around 1200 observations. Second, the loss functions of the Chinese stock and WTI oil markets show a U-shape. The model's loss functions tend to decrease and then increase as window size increases. The DM test is significant at the 10% level in the parts with lower loss functions. When forecasting the volatility of the Chinese stock market and WTI oil market, the window sizes should be set at roughly 1500–2000 observations and roughly 1500 observations, respectively. The aforementioned results suggest that a window size of 1000–2000 observations is ideal for minimizing forecast errors across different assets. Then we conduct an economic value test. It turns out that the utility of the model is usually larger when the loss function is lower. That means it can bring economic gains to investors. Finally, our results pass a series of robustness tests.

Acknowledgments

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

[1] https://realized.oxford-man.ox.ac.uk/

[2] For space reasons, we only report the DM test results for the HAR-RV-size model, with the window size increasing by 100 observations at a time.

[3] For space reasons, we only report the results of portfolio performance with the window size increasing by 100 observations at a time.

[4] We conduct the DM tests of all robustness analyses, the results of which are available upon request.

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